

LAB MANUAL
Experiment No.: 03
Unsupervised Learning for Process Optimization using K-Means and Hierarchical Clustering

Aim

To apply K-Means and Hierarchical Clustering on an **open-source manufacturing dataset** to group similar machine operations and identify possible anomalies.

Objectives

- To understand unsupervised learning.
- To perform K-Means clustering.
- To perform Hierarchical clustering.
- To analyze clusters for process optimization.

Software Required

- Python
- Jupyter Notebook / Google Colab
- Python Libraries:
 - pandas
 - numpy
 - matplotlib
 - scikit-learn

Dataset

Use any suitable **open-source manufacturing dataset** (e.g., Kaggle, UCI Machine Learning Repository, or other public repositories).

Dataset Name:

Source:

Parameters Used:

1. _____
2. _____
3. _____
4. _____
5. .
6. .
7. .

Theory

Unsupervised Learning

Unsupervised learning identifies patterns or groups in data without predefined labels.

K-Means Clustering

K-Means divides data into a specified number of clusters based on similarity.

Hierarchical Clustering

Hierarchical clustering creates clusters in a tree-like structure (dendrogram) showing relationships among data points.

Procedure

1. Download an open-source manufacturing dataset.
2. Import the required Python libraries.
3. Load the dataset.
4. Select the required machine parameters.
5. Perform data preprocessing (if required).
6. Apply K-Means clustering.
7. Visualize the clusters.
8. Apply Hierarchical clustering.
9. Plot the dendrogram.
10. Analyze the obtained clusters and identify possible anomalies.

Observations

Dataset Details

Parameter	Value
Number of Samples	___168___
Number of Features	____7____
Selected Features	mean_voltage, min_voltage, max_voltage, mean_current, mean_temperature, max_temperature, discharge_time

K-Means Clustering

Cluster Number	Number of Data Points	Remarks
Cluster 1	54	Similar battery operating conditions
Cluster 2	83	Similar battery operating conditions
Cluster 3	31	Similar battery operating conditions

Hierarchical Clustering

Observation	Remarks
Number of major clusters	3
Any outliers observed	Yes, a few distant observations may be observed
Interpretation	The dendrogram shows grouping of battery cycles having similar voltage, current, temperature and discharge characteristics

Results

Code:-

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans
from sklearn.decomposition import PCA
from scipy.cluster.hierarchy import dendrogram, linkage
df=pd.read_csv("/media/galdruh/galdruh_storage/sem_5/AIR/dataset/batter
y_features.csv")
```

```

features=["mean_voltage","min_voltage","max_voltage","mean_current","mean_temperature","max_temperature","discharge_time"]
X=df[features].dropna()
scaler=StandardScaler()
X_scaled=scaler.fit_transform(X)
inertia=[]
for k in range(2,9):
    model=KMeans(n_clusters=k,random_state=42,n_init=10)
    model.fit(X_scaled)
    inertia.append(model.inertia_)
plt.figure(figsize=(8,5))
plt.plot(range(2,9),inertia,marker="o")
plt.xlabel("Number of Clusters")
plt.ylabel("Within-Cluster Sum of Squares")
plt.title("Elbow Method for K-Means")
plt.grid(True)
plt.show()
kmeans=KMeans(n_clusters=3,random_state=42,n_init=10)
clusters=kmeans.fit_predict(X_scaled)
X_result=X.copy()
X_result["Cluster"]=clusters
pca=PCA(n_components=2)
X_pca=pca.fit_transform(X_scaled)
plt.figure(figsize=(8,6))
plt.scatter(X_pca[:,0],X_pca[:,1],c=clusters,s=40)
plt.xlabel("Principal Component 1")
plt.ylabel("Principal Component 2")
plt.title("K-Means Clustering")
plt.grid(True)
plt.show()
cluster_counts=X_result["Cluster"].value_counts().sort_index()
print("K-Means Cluster Counts:")
print(cluster_counts)
cluster_centers=pd.DataFrame(scaler.inverse_transform(kmeans.cluster_centers_),columns=features)
print("\nCluster Centers:")
print(cluster_centers)
distances=np.linalg.norm(X_scaled-kmeans.cluster_centers_[clusters],axis=1)
threshold=np.percentile(distances,95)
X_result["Distance"]=distances
X_result["Anomaly"]=X_result["Distance"]>threshold
print("\nAnomaly Threshold:",threshold)

```

```
print("Number of Possible Anomalies:", X_result["Anomaly"].sum())
print("\nPossible Anomalies:")
print(X_result[X_result["Anomaly"]][features+["Cluster", "Distance"]])
sample_size=min(169,len(X_scaled))
sample_indices=np.random.RandomState(42).choice(len(X_scaled),sample_size,replace=False)
X_sample=X_scaled[sample_indices]
linked=linkage(X_sample,method="ward")
plt.figure(figsize=(14,7))
dendrogram(linked,no_labels=True)
plt.xlabel("Data Points")
plt.ylabel("Euclidean Distance")
plt.title("Hierarchical Clustering Dendrogram")
plt.show()
print("\nHierarchical Clustering completed successfully.")
print("Number of samples:",len(X))
print("Number of clustering features:",len(features))
print("Number of major clusters considered:",3)
```

Output:-

```

C:\venv\gator>gator-ux -displotm -5503:/media/gator-ux/gator-ux-storage/ser_3/uk/exp_35 - python3 exp3.py
K-Means Cluster Counts:
Cluster
0      54
1      83
2      31
Name: count, dtype: int64

Cluster Centers:
   mean_voltage  min_voltage  max_voltage  mean_current  mean_temperature  max_temperature  discharge_time
0      3.540863    2.660768    4.199616    -1.883963         32.187085         39.036598        3234.280463
1      3.487371    2.666944    4.196368    -1.732538         33.395113         40.854549        2919.595542
2      3.553609    2.580125    4.188622    -1.872698         32.227243         38.550793        3509.110903

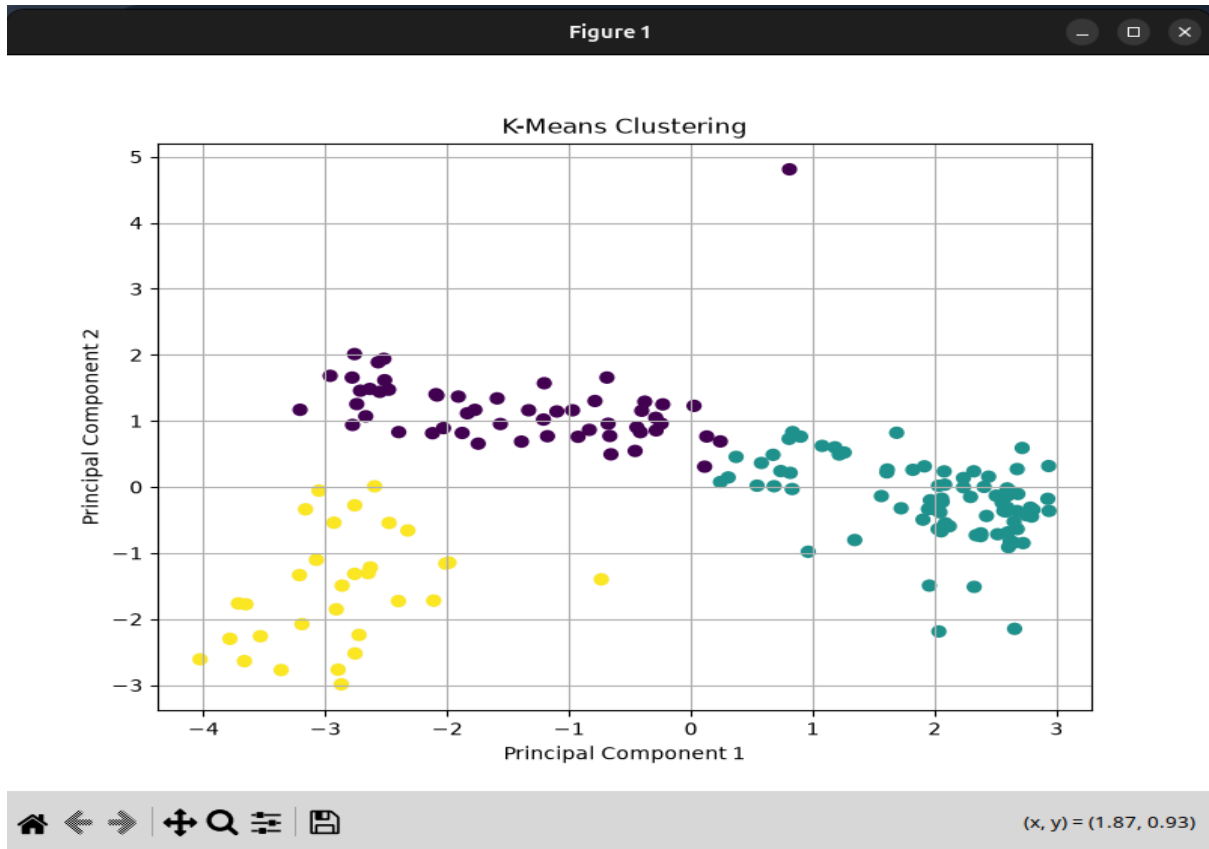
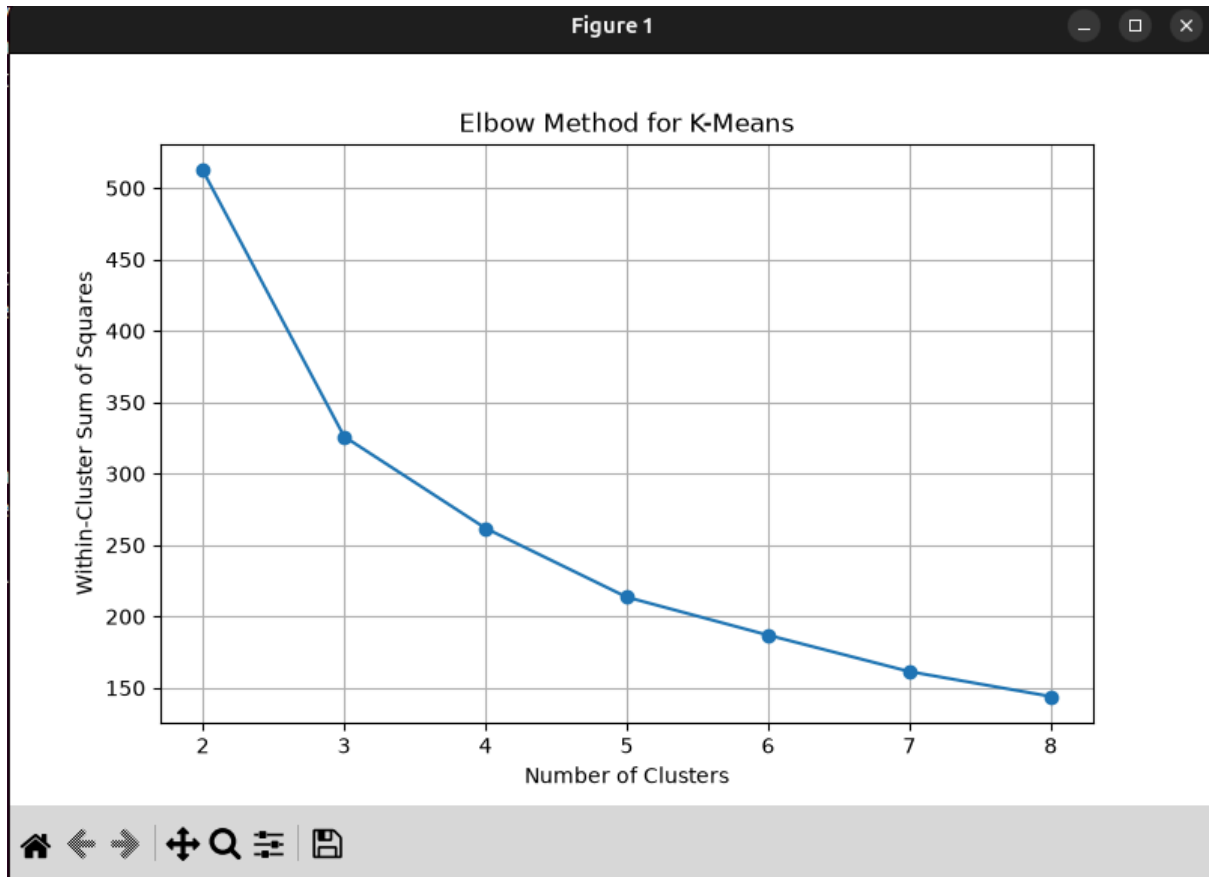
Anomaly Threshold: 2.3260909048665277
Number of Possible Anomalies: 9

Possible Anomalies:
   mean_voltage  min_voltage  max_voltage  mean_current  mean_temperature  max_temperature  discharge_time  Cluster  Distance
14      3.562747    2.694622    4.189110    -1.881794         32.044070         38.213586        3450.297      2  2.396374
18      3.557887    2.455679    4.189273    -1.913160         31.966242         38.546590        3390.281      2  2.654253
22      3.561410    2.690470    4.187344    -1.883194         31.987958         38.165816        3492.907      2  2.399770
25      3.564103    2.684999    4.188261    -1.903053         31.688747         37.982161        3431.125      2  2.392727
27      3.556833    2.471315    4.187560    -1.913621         31.565586         38.188311        3410.688      2  2.515556
47      3.559765    2.605158    4.190409    -1.938881         34.310885         40.712182        3328.937      2  3.732029
89      3.522222    2.692055    4.222920    -1.822917         32.856296         40.094181        3177.219      0  5.280837
149     3.467524    2.664632    4.184535    -1.686381         33.037928         40.661842        2838.265      1  2.730281
166     3.471071    2.678714    4.184827    -1.688898         33.713519         41.047030        2802.016      1  2.667049

Hierarchical Clustering completed successfully.
Number of samples: 168
Number of clustering features: 7
Number of major clusters considered: 3

```

K-Means Cluster Plot:-



Hierarchical Dendrogram:-

